

Algebra PhD Exam, January 2001

Time allowed: 240 minutes. Do seven of the following eleven problems. Please do not turn in more than seven problems. You must show your work. Answers with no work and/or no explanations will receive no credit. State clearly any theorem you use in your proofs. In the problems, $\mathbb{Z}$, resp. $\mathbb{Q}$, $\mathbb{C}$, is the set of all integers, resp. of all rational numbers, of all complex numbers.

1. Let G be a finite simple (abelian or non-abelian) group of order n . Find the number of normal subgroups of $G \times G$.
2. Show that there is no simple group of order 112. (Caution: You are not allowed to use Burnside's $p^a q^b$ -Theorem.)
3. Let N be a minimal normal subgroup of a finite group G . Assume N is solvable. Show that N is an elementary abelian p -group for some prime p .
4. Let A, B be two matrices over a field F . Suppose that $\begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix}$ is similar to $\begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix}$. Prove that A is similar to B .
5. Determine the Galois group over $\mathbb{Q}$ of
 - a) $(x^3 - 2)(x^2 - 3)$;
 - b) $(x^3 - 2)(x^2 + 3)$.
6. Is there any integer d such that: for any separable extension F of degree 5 of a field K , if an extension E of K is a normal closure of F/K then $[E : K] \leq d$? If yes, prove your bound. If not, justify why.
7. Let $\xi \in \mathbb{C}$ be such that $\xi^{2000} = 3$. Show that -3 is not a sum of squares in $\mathbb{Q}(\xi)$.
8. Prove the Hilbert Basis Theorem: If R is a commutative Noetherian ring with identity, then so is $R[x_1, x_2, \dots, x_n]$.
9. Let A be a finite dimensional (not necessarily commutative) $\mathbb{C}$ -algebra with no zero divisors, but with identity. Show that $\dim_{\mathbb{C}} A = 1$.
10. This problem has two parts.
 - a) If the ring R is a principal ideal domain, prove that every nonzero prime ideal in R is maximal.
 - b) Is the statement in a) still true if R is not a principal ideal domain? (Hint: Consider the ring $\mathbb{Z}[x]$.)
11. Suppose that R is a ring with identity.
 - a) Prove that each free left R -module is projective.
 - b) Prove that a left R -module P is projective if and only if each short exact sequence of left R -modules

$$0 \longrightarrow A \longrightarrow B \longrightarrow P \longrightarrow 0$$

splits.